AXT (AXTI)

Full investment research | Model assumptions and source evidence

Public figures use whole-number presentation. Exact calculations, share counts, operating costs and every probability control remain in the full model JSON. Original source graphics retain their original units and precision. **CURRENT EDITION - 19 October 2026 | Framework 8.0**

Full investment research, current probability-weighted operating model and analytical rescore. Financial assumptions, source evidence and forward estimates are distinguished throughout.

Q4 2027 ceiling

Current model item	Value	
Q4 2027 ceiling		68 USD/share
Valuation window	Q4 2027 to Q3 2028	
Artifact analytical Base		75/100
Protected portfolio Excel Base		75/100

A 30% allowed-route price outcome is offset by a 10% restrictive-origin outcome. The weighted result is slightly negative after stranded manufacturing costs and parent ownership.

The sanctioned valuation remains **20x forward-four-quarter operating profit**, with the existing accounting basis, ownership, shares and currency conversion. The active ceiling next rolls on January 1, 2027.

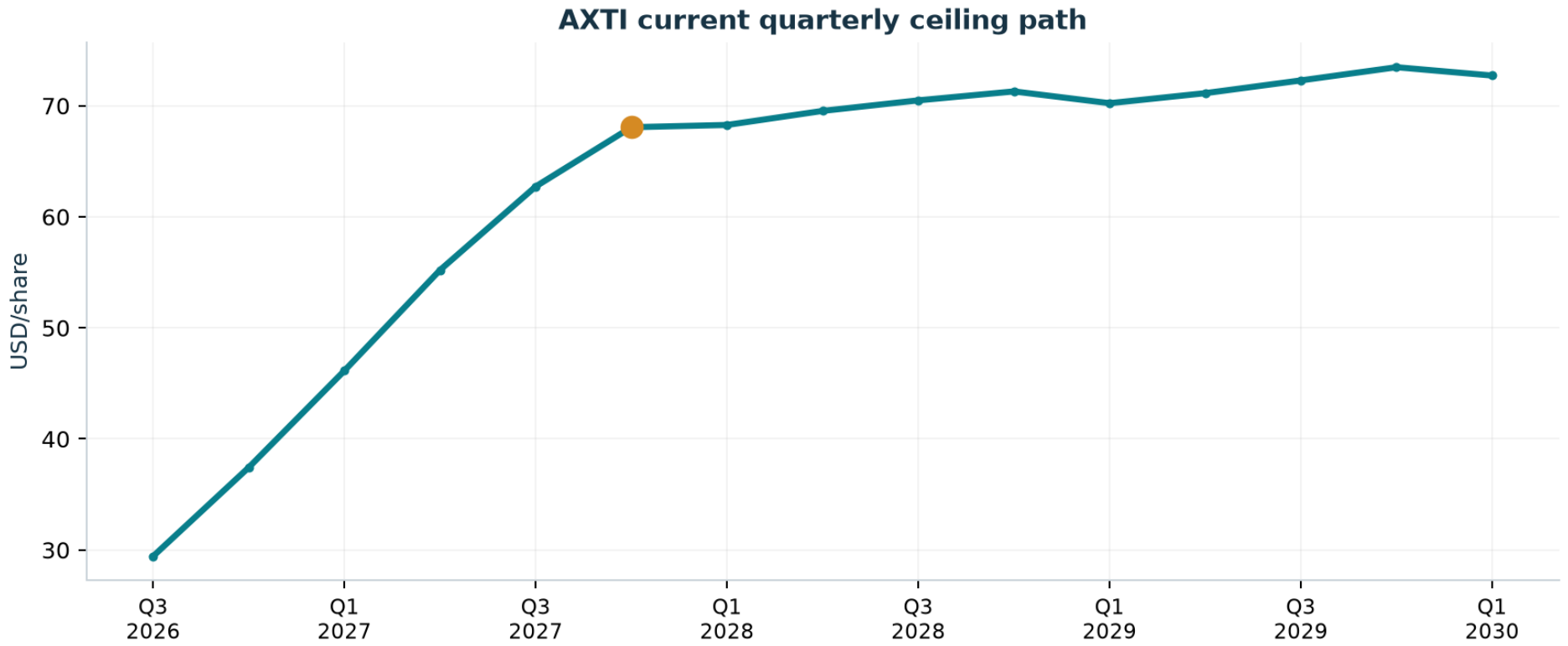


Figure 1. Current quarterly ceilings, USD/share. Gold marks the active Q4 2027 horizon. The 40% policy contribution is included once in operating earnings.

Ceiling quarter	Current ceiling USD/share
Q3 2026	29
Q4 2026	37
Q1 2027	46
Q2 2027	55
Q3 2027	63
Q4 2027	68
Q1 2028	68
Q2 2028	70

Ceiling quarter	Current ceiling USD/share
Q3 2028	70
Q4 2028	71
Q1 2029	70
Q2 2029	71
Q3 2029	72
Q4 2029	73
Q1 2030	73

Analytical Base and scored conviction

Category	Points / maximum	Current review reason
Core thesis	17/20	Qualified low-defect crystal production, integrated raw materials and customer reservations retain a moat. Low six-inch finishing yields demonstrate difficulty; competitors and downgraded material prevent a monopoly inference.
Cycle position	9/10	Rising current demand and 2027 contract delivery remain the inflection. The source does not establish peak customer consumption; stockpiles and 2028 excess supply remain material counters.
Forward TAM	9/10	Optics power and die-area intensity support substrate demand already in the expansion path. No additional multiplier from CPO source counts or mixed wafer equivalents.
Catalysts	5/5	Q3 results, 2026 exit capacity, signed 2027 deliveries and six-inch customer milestones remain concrete. New report supplies no new dated AXT contract.
Macro and policy	7/10	AI optics demand remains strong; Chinese export permits and downstream origin policy remain separate constraints.
Alpha and evidence	3/5	Diameter, contract economics and parent capture remain the useful analytical edge. Anonymous supply figures do not improve company unit/margin visibility enough to raise the score.
Risk deduction	-5	Retain five classes: permits/geopolitics; yield/capex execution; reservation/refund and double-order risk; competition/substitution; ownership/redemption/dilution.

Analytical Base 45/60 = 75; protected Excel Base 75. The score is independent of share price and does not multiply valuation.

The existing policy probability weights only its incremental revenue, costs and earnings. The current source review below reassesses all seven score categories without changing protected portfolio conviction or the selected financial case.

October 19 current source review and full rescore

The complete new InP source was reviewed, the seven conviction categories were reassessed, and the full affected financial path was compared with the current master. All numerical operating inputs, valuation methods, shares and protected Excel Base values are retained.

Source identity, product stages and contrary views

P Equity Research published the 16-page InP Expert Call on October 19, 2026. The call is dated October 2026 without an exact day. The expert is described as China-based; name, employer and credentials are not supplied. The author's commentary and expert's views are separate evidence, rather than two independent interviews. No source-authored probability is provided. The original is retained with SHA-256 b457250e01dc31744a15836b260540bfdaf1348dcb9862e6cc2c096bf2b98ddc.

The expert expects capacity to meet demand around H2 2027, some 2027 price easing and clearer oversupply in 2028. The publisher disagrees and cites qualification and yield constraints that could prolong tightness through 2028. Both views remain visible. The expert's three-inch-equivalent totals and Figure 25's four-inch-equivalent chart have an unresolved axis multiplier; neither is converted into company sales. The claim that high-power chips require larger die area and lower defects describes resource intensity, with no universal supplier-volume multiplier.

Fewer than 10 of 100 grown crystals reportedly become finished six-inch wafers; much of the remainder is downgraded into saleable three- or four-inch products. This is a diameter/grade finishing ratio, not 90% material scrap, laser-die yield or module good-output yield. Customer qualification, export permission, substrate finishing, epitaxy, device processing and module acceptance remain distinct gates. Fixed-price and fixed-volume long-term contracts do not automatically reset after spot quotes change.

The interview describes roughly 5% recent price gains, diameter-specific domestic quotes and a 10-20% overseas contract premium. Grade, contract date, processing and customer terms differ. Its 18-CW-source CPO example belongs to a particular architecture. The raw-phosphorus and long-MOCVD-lead-time claims do not identify a named company allocation, delivery failure or incremental invoice. They are not mapped to every optical supplier.

Company profit capture, overlap and bounded proxies

AXT's existing InP reference exit revenue capacity rises from $60 million to $130 million between 2026 and 2027. Core and six-inch product vectors already separate mix and margin. Generic output doubling has 100% directional overlap with that ramp; the new source supplies no diameter-matched accepted-unit schedule above it. Core InP gross margin rises from about 52% in Q3 2026 to 58% in Q4 2027, while six-inch gross margin reaches about 63%. The six-inch premium still requires company-specific grade recovery and yield evidence.

The price diagnostic assigns 15% probability to a conditional 5% price increase on 20% repriceable InP receipts. It charges 5% of extra revenue for manufacturing/support and another 5% for operating costs, then applies the existing parent economic share once. Before overlap, Q4 2027's four-quarter parent operating-profit contribution is about $658,000, or less than $1 per share. A separate grade-cost stress assigns 15% probability to 20% extra manufacturing cost on six-inch revenue above the Q4 2026 $10 million base and deducts 50% overlap with existing ramp costs. It produces a four-quarter parent-profit reduction of about $754,000, also less than $1 per share. These probabilities and conditions are explicit analytical diagnostics, not source facts or additional selected branches.

The selected new numerical increment is zero. Existing product margins, the explicit 5% premium on the 20% uncommitted policy cohort, grade execution and later margin normalization already capture the mapped mechanisms. A second premium has no separate contract/grade cohort; a new cost shock has no measured salvage recovery or current cost allocation. The low six-inch finishing percentage does not supply that missing cost. The unchanged result follows specific overlap and unit scope, independent of publication thresholds or the absence of official confirmation.

Issuer contracts, permits and current counter-evidence

AXT's July 30 Q2 release reported about $48 million revenue and 45% gross margin, with record InP revenue. It did not disclose a realized wafer ASP, shipment count or the expert's 2027 physical-unit forecast. The Q2 filing identifies Casela's fixed 2027 reservation and prepayment, Coherent's three-year six-inch capacity reservation and prepayment/refund terms, and Lumentum's later six-year reservation. Prepayments fund execution and introduce delivery/refund obligations; they are not extra product sales or fresh margin windfalls. Contracts already reflected in the model are counted once. Chinese export licenses and downstream origin restrictions remain separate risks.

Next evidence: accepted output by diameter and grade; value recovered on downgraded wafers; comparable contract renewal prices; actual new-line cost absorption; permit conversion; and reservation deliveries/refunds. A missed customer qualification, buffer liquidation or fixed-cost underabsorption can weaken the ramp. Qualified new capacity and sustained like-for-like realization can strengthen it.

Primary sources: Q2 release and Q2 SEC filing. Issuer and SEC pages were read through supported web access; direct original-download requests returned 403, so no locally saved full-original hash is claimed.

Original source pages

Q2. How is the InP substrate consumption of a single optical module calculated?

Optical modules are mainly divided into two technical routes, EML and CW. Taking a 3-inch substrate as an example, without considering downstream epitaxy and optical chip yield, one substrate can be cut into about 10,000 EML chips for 100G and 200G. But once downstream yield is included, the effective chip output per wafer is a little over 2,000. For conventional 70 mW and 100 mW CW chips, one 3-inch substrate can be cut into about 8,000 dies, and after yield the effective output is likewise a little over 2,000. Although CW chips have fewer dies per wafer, their overall yield is usually slightly higher than EML's.

Q3. How will CPO solutions affect demand for InP substrates?

CPO is currently at the sample stage and not yet in large-scale volume production. CPO will drive significant growth in InP substrate demand, mainly through die area and quality requirements. **A CPO module is**

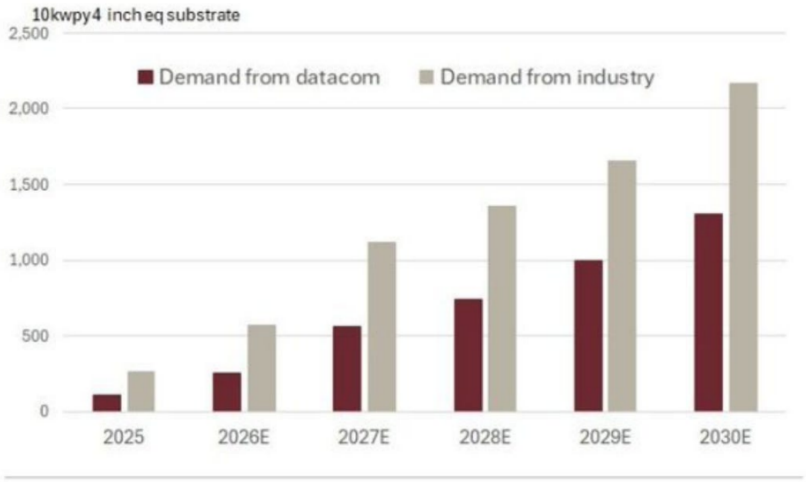
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expected to use 18 CW light sources, with power requirements far above conventional products. For example, the 200 mW and 300 mW CW chips being developed by Yuanjie Semiconductor have a single bare die area 2 to 3 times that of today's 70 mW and 100 mW chips, which directly increases substrate area demand. CPO modules also place stricter demands on substrate quality. Because continuous emission from CW lasers generates serious heat, high defect density raises the probability of chip failure. Therefore, for 300 mW CW sources used in CPO, **the maximum substrate defect density is required not to exceed 200**, whereas for orders for conventional 100G and 200G optical modules or 70 mW and 100 mW CW sources, **the average defect density requirement is only below 500**.

Q4. How will market supply meet the demand growth expected in 2027?

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Figure 25: InP substrates demand forecast



Source: Yole,Company data, UBSe

Q6. What is your view of the capacity expansion plans of major

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manufacturers such as AXT and Yunnan Germanium, and their effect on market supply next year (2027)?

The capacity targets of 200,000 or 300,000 wafers mentioned by AXT and Yunnan Germanium should be long-term plans. By 2027, the limit of their capacity expansion is roughly double 2026 levels. Even so, AXT's monthly capacity could then reach nearly 100,000 wafers, with annual capacity above 1 million. Yunnan Germanium and Xiandao will not be small either; together the three are expected to achieve annual capacity of more than 3 million wafers in 2027. Therefore, by the second half of 2027, the tight market supply is expected to ease.

Q7. How have InP substrate prices changed recently, and how should future price trends be judged?

Prices have risen slightly recently, by about 5%, but far less than the rise in the first half of 2026. Specifically, 2-inch products were around RMB 1,500 last month (September 2026); 3-inch products are close to RMB

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Policy breadcrumb and economic hypothesis

Serenity's October 1 post attributes Washington meetings to Morgan Stanley and reports a possible 3.2T-focused restriction, with a potential 65% U.S.-value route for Chinese modules. The source expects laser pricing to matter more than broad module-share redistribution. Treat this as an attributed policy scenario; the original MS note and final FCC language establishing these exact terms were not recovered. The separate September 25 Senate proposal concerns sensitive federal systems and a five-year transition. It is not confirmation of these FCC terms.

Serenity policy post | Foundry follow-up | Separate Senate proposal

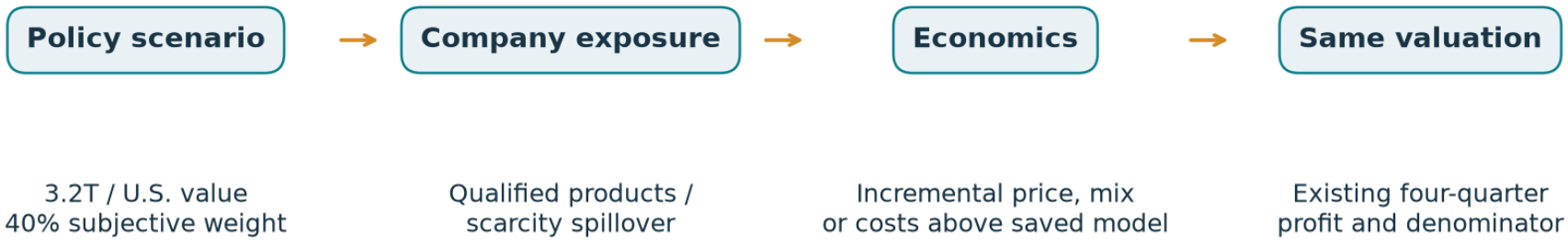


Figure 2. Analytical transmission of the attributed policy scenario. This is a model diagram, not an official regulatory diagram or customer award.

Assigned common policy-scenario probability: 40%. Subjective initial weight above the single-source 25-35% band because Serenity has the owner's highest weighting and attributes Washington meetings to Morgan Stanley. The exact future rule, timing and economic effect remain uncertain. Serenity and Morgan Stanley are one source chain, not two independent confirmations.

Conditional on policy, 75% weight allows the upstream substrate route and a 5% repricing of 20% uncommitted InP sales; 25% instead tests a distinct restrictive-origin result losing 10% InP volume, with 10% of displaced COGS stranded. Branches are mutually exclusive and sum to 100%. Neither branch is a claim about enacted rules. Existing permit risk and planned capacity remain intact. Parent ownership applies once to incremental OP.

Eligible fractions are bounded scenario assumptions, not origin certificates. U.S. corporate domicile and a U.S. listing do not prove manufacturing origin. If the final content definition excludes the modeled products, their eligible fraction goes to zero.

For component and integrated suppliers, the filtered revenue pool means 3.2T-linked sales plus incremental cross-generation pricing spillover from competition for common qualified capacity. It is not a claim that 800G/1.6T is directly regulated. Fractions are LOW-confidence exposure proxies, not verified geographic market shares.

Signal ledger field	Current treatment
Signal ID / date	FCC-OPTICS-20261001; Serenity/MS source chain
What is already included	All saved product growth, shortages, capacity, customer probabilities and contracts. The bridge adds only the explicitly calculated incremental economics.
Probability and timing	40% common policy scenario; price/cost intensity 0 before Q3 2027, 25% Q3 2027, 50% Q4 2027, 75% Q1 2028, 100% from Q2 2028. AAOI mix uses its own slot ramp.
Numerical contribution	Q4 2027 operating quarter: delta revenue negative under 1, delta consolidated OP negative under 1 (USD millions). Forward-four-quarter valuation OP changes -1.
Dependence	The policy probability is shared across companies. These are correlated portfolio exposures, not independent confirmations. Existing customer-conversion probabilities are already inside source revenue.
Raise weight	Rule/proposal scope supports qualifying content and timing; company product/foundry evidence improves; customer sourcing or like-for-like price negotiations support the incremental premium.
Reduce weight	Policy is delayed beyond the horizon, the content definition excludes products, supply eases, fixed contracts prevent repricing, unit costs absorb the premium, or 3.2T conversion slips.
Zero-contribution boundary	No broad TAM uplift, no new factory capacity, no OCS benefit and no blanket-ban recovery. AAOI receives no modeled units before 2028. MXL is assessed separately as a candidate.

Conditional inputs selected before calculating targets

Input	Value	Evidence status
Allowed route conditional probability	75%	LOW-confidence analyst assumption; inherited cap where labeled
Uncommitted InP repricing fraction	20%	LOW-confidence analyst assumption; inherited cap where labeled
Conditional incremental price premium	5%	LOW-confidence analyst assumption; inherited cap where labeled
Allowed-branch incremental COGS rate	5%	LOW-confidence analyst assumption; inherited cap where labeled
Allowed-branch incremental opex rate	5%	LOW-confidence analyst assumption; inherited cap where labeled
Restrictive route conditional probability	25%	LOW-confidence analyst assumption; inherited cap where labeled
Restrictive-branch InP volume loss	10%	LOW-confidence analyst assumption; inherited cap where labeled
Displaced product COGS that remains fixed	10%	LOW-confidence analyst assumption; inherited cap where labeled

Incremental COGS and opex rates apply to incremental price revenue, not original product revenue or COGS. AAOI mix cost uses unit COGS separately. AXTI stranded COGS is a fraction of COGS attached to lost units. Chinese component-cost rates apply to affected reference COGS.

Timing is an analyst sensitivity. A deliberately labeled early-sourcing proxy ahead of a 2028 3.2T ramp. Earlier quarters receive zero incremental policy revenue. Full intensity begins Q2 2028 and remains an uncertain continuation through the saved forecast only. These dates are not announced orders or a published regulatory timetable. The probability sensitivity below includes a no-policy control; a later start would reduce the near-term contribution.

Mutually exclusive origin outcomes

Outcome	Unconditional probability	Economics
Reference / no incremental policy effect	60%	Existing shortage, capacity and permit assumptions
Allowed upstream route	30%	5% premium on 20% of InP sales; 5% incremental COGS and 5% opex
Restrictive origin interpretation	10%	10% InP volume displacement; 90% of attached COGS saved, 10% stranded; no fixed opex saving

The restrictive branch is an analyst risk sensitivity, not an assertion that AXT substrates are covered or disqualified. Substrate value could fit inside a permitted non-U.S. share, or later processing could determine origin. Conversely, a different rule definition could impair sales. Existing Chinese export-permit risk is not deducted again. The 86% parent economic ownership applies once after incremental consolidated operating profit.

Incremental revenue, cost and profit

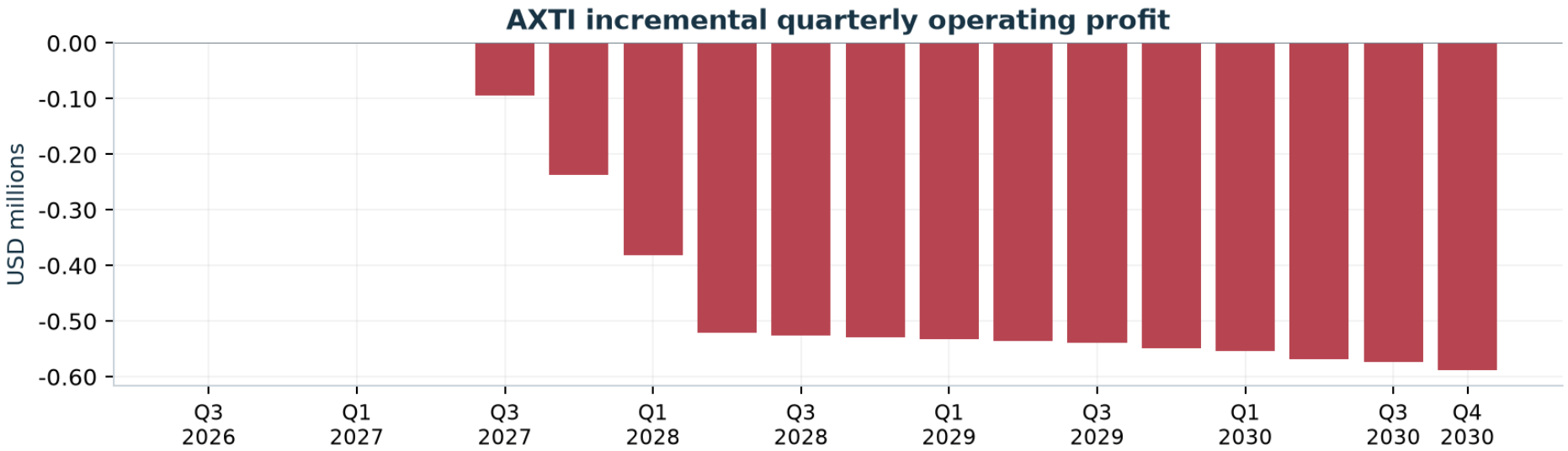


Figure 3. Incremental consolidated operating profit, USD millions. Negative bars represent the net weighted adverse policy effect; baseline operations are unchanged.

Quarter	Delta revenue, USD thousands	Delta COGS, USD thousands	Delta gross profit, USD thousands	Delta opex, USD thousands	Delta OP, USD thousands
Q3 2026	0	0	0	0	0
Q4 2026	0	0	0	0	0
Q1 2027	0	0	0	0	0
Q2 2027	0	0	0	0	0
Q3 2027	-184	-93	-91	4	-95
Q4 2027	-455	-228	-227	10	-236
Q1 2028	-735	-370	-365	16	-381
Q2 2028	-1,036	-537	-499	22	-521

Quarter	Delta revenue, USD thousands	Delta COGS, USD thousands	Delta gross profit, USD thousands	Delta opex, USD thousands	Delta OP, USD thousands
Q3 2028	-1,085	-582	-503	23	-527
Q4 2028	-1,134	-629	-505	24	-529
Q1 2029	-1,176	-669	-507	25	-533
Q2 2029	-1,218	-708	-510	26	-536
Q3 2029	-1,260	-748	-512	27	-539
Q4 2029	-1,302	-782	-520	28	-548
Q1 2030	-1,337	-812	-525	29	-553
Q2 2030	-1,372	-834	-538	29	-568
Q3 2030	-1,407	-864	-543	30	-573
Q4 2030	-1,442	-885	-557	31	-588

Active-window product bridge

Product / event	Pre-policy operating pool, USD thousands	Filtered exposure, USD thousands	Weighted delta revenue, USD thousands	Weighted delta OP, USD thousands
Core InP: 2-, 3- and 4-inch substrates allowed route	91,000	18,200	136	123
Core InP: 2-, 3- and 4-inch substrates restrictive route	91,000	9,100	-455	-282
6-inch InP and emerging optical architectures allowed route	39,000	7,800	58	53
6-inch InP and emerging optical architectures restrictive route	39,000	3,900	-195	-130

Filtered exposure is a scenario allocation, not verified U.S.-origin revenue. For cost cases the exposure is an affected cost pool; AAOI substitution units are explained in its separate bridge. All financial amounts use the company table units.

Business research and source evidence

The following sets out the company thesis, historical results, guidance, capacity, program probabilities, risks and original source images. Source dates remain visible. Operating assumptions below feed the complete current financial tables, including the explicit probability-weighted policy contribution.

1. Supercycle, supply chain and durable advantage

AXT grows compound-semiconductor crystals and supplies substrates used to manufacture lasers and detectors. More high-speed optical links and larger, lower-defect wafers increase its addressable content. In-house furnace design, material supply and qualification support expansion, while all current manufacturing remains exposed to Chinese export permits. Competitor capacity and eventual substitution matter; being upstream of optics does not mean earning downstream laser or transceiver revenue.

Indium/phosphorus feedstock → crystal growth → slicing/polishing and defect control → epitaxy/device manufacturing → optical engine/transceiver → switch/server. AXT operates at the substrate and selected raw-material layers. The 6-inch wafer provides 2 times the geometric area of a 4-inch wafer before edge loss; this is not a guarantee of 2 times die yield or dollar margin. Customers still need qualified epitaxy and device processes.

Business / curve	Stage	Evidence and next inflection
800G/1.6T substrate demand	Scaling, 25-50% qualitative band	Record InP sales and expanding revenue-capacity targets.
6-inch InP	Early, below 25% band	Customer qualification and supply agreements; exact sales undisclosed.
CPO/NPO scale-up	Early, below 25% band	Potential later-2027+ driver, dependent on architecture adoption.
GaAs and raw materials	Established; selective growth	VCSEL, RF, robotics and internal supply assurance differ from InP scarcity.

Stage percentages are qualitative framework bands relative to a possible mature opportunity, not measured market penetration. A current capacity constraint can coexist with a larger long-term TAM; qualification and economic capture decide how much reaches this company.

2. Forward TAM and the three-phase expansion lens

Phase	Engineering opportunity	Demand-to-company bridge
2026-27 / 100% thematic weight	800G/1.6T laser and detector substrate growth	66M total Q3 guide and 60M/130M InP exit revenue-capacity targets calibrate feasible near-term capture.
2028-29 / 50% thematic weight	More optical links, larger wafers and NPO/CPO	Management relayed customer/industry expectations of 4-6× substrate-market growth over three to five years. This is attributed forward evidence, not an independently audited TAM.
2030+ / 20% thematic weight	Broader scale-up optics and high-quality material content	Reference InP reaches 794M in 2030 before the current model policy increment; exact market size, wafer-equivalent units, ASP and share remain an open engineering check.

The 100% / 50% / 20% thematic weights guide thesis assessment. They are not blanket discounts to reported sales, contracts or all forecasts. Programme probabilities below govern only separately identified optionality. Current reported sales calibrate the starting point; they do not cap the future market.

3. Current earnings and management evidence

Q2 revenue was 47,589,000, GAAP GP 21,366,000, R&D 3,651,000 and SG&A 7,292,000: OP 10,423,000. Q1 revenue was 26,924,000 and OP -1,585,000. Q2 rounded category disclosure was InP 30,700,000, GaAs 6,600,000, Ge 272,000; the filing reports raw materials/other 10,026,000. Reconcile rounded categories by assigning InP 30,691,000. Q3 management indicated 66M permit-covered or domestic revenue, around 11M GAAP opex and 66,500,000 diluted shares. The model uses those levels, rather than the old 59,500,000/64M assumptions. [S1/S2/S4]

April 30 and July 30 displayed calls were reviewed through Q&A and closing. The earlier call targeted about 35M quarterly InP revenue capacity exiting 2026 and 65-70M exiting 2027/early 2028. July increased those capacity estimates to about 60M and 130M, citing throughput, diameter/mix and physical additions. Those are capacity estimates, not booked quarterly revenue. The CFO suggested near-current GM for immediate forecasts while describing an eventual 50%+ aspiration. The model separates this management evidence from an estimated diameter-margin premium. [S3/S4]

Both displayed calls were reviewed through closing; the host nevertheless labels access a preview, so fidelity against original audio is not certified. The July transcript reverses customer prepayment amounts in one passage; SEC contractual terms control. No accessible subsequent September conference transcript or revised formal guide was verified. Exact global capacity shares, realized wafer ASPs and product gross margins remain undisclosed.

4. Local evidence and calibration packet

The old very small published substrate-market totals do not reconcile with current company capacity plans; neither those totals nor laser-industry revenue can be adopted as substrate TAM without a unit/ASP bridge. AH12 uses customer procurement directly: Casela's 2027 purchase amount and Coherent/Lumentum capacity deposits support real demand, but the latter deposits are not total contracts or recognized sales. Management reaffirmed roughly 30-40M 2026 capex and around 100M in 2027; a possible later greenfield facility was discussed at 220-250M. H1 actual operating cash flow was -900,000 and cash PP&E purchases 8,598,000, so cash generation has not yet caught up with the planned expansion. Restricted cash is not freely available parent liquidity. [S2/S3/S4]

6. Revenue sources, capture and product margins

Preserve five distinct revenue/margin sources and compute group GP from them. Update the current-quarter guide and share count, shift more of early InP revenue toward 3/4-inch, and explicitly allocate the Tongmei minority claim. The 6-inch revenue split is not reported and should never be labeled actual. Consolidated raw-material revenue excludes equity-accounted affiliate sales and eliminates internal substrate-material transfers.

Vector	Actual calibration / source	Retained product and accounting caveats
Core InP	Estimated 28M of reconciled Q2 InP total 30,691,000	Capacity conversion and permits, not total laser TAM, drive units. Estimated, not disclosed historical product GM.
6-inch InP	Estimated 2,691,000 Q2 residual; could include qualification material	Shifts some early revenue back to established diameters versus the original. Qualification can erase this premium.
GaAs	6,600,000 rounded Q2 actual	VCSEL is a partial demand hedge, not a guaranteed offset. Competitive industrial/RF and optical mix.

Vector	Actual calibration / source	Retained product and accounting caveats
Germanium	272,000 Q2 actual	Small contribution.
Raw materials and other	10,026,000 filing actual	Consolidated external sales only. Materials cycle and customer mix. Further lower-tier minority claims are a remaining proxy limitation.

Confidence: reported historical totals are HIGH. Source-based forward extrapolations are MEDIUM only where the cited source sizes the relevant period; other product allocations, future capture and margin assumptions are LOW. A source for the historical anchor does not make an unsourced future factor management guidance. The inherited factors describe the reference model; the current chapter supplies revised forecast calculations.

V1. Core InP: 2-, 3- and 4-inch substrates

Sources: S2/S4 category totals and capacity. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

V2. 6-inch InP and emerging optical architectures

Sources: S3/S4/S5 diameter qualification; modeled split. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

V3. Gallium arsenide substrates

Sources: S4 category disclosure and VCSEL applications. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

V4. Germanium substrates

Sources: S4 category disclosure. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

V5. Consolidated raw-material companies and other

Sources: S2 reported raw-material/other category. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

7. Operating-cost bridge and quarterly operating model

Inherited reference model. Use 11M GAAP opex in Q3, matching management guidance, then 13,500,000 in Q4 and $15 million / $16,500,000 / $18 million / $20 million across 2027. It rises progressively to 32,500,000 per quarter by Q4 2030; ratios are calculated from those amounts. This retains SBC, engineering, selling and administration. Depreciation and plant start-up costs are embedded in product margins. Opex grows in absolute dollars but much more slowly than revenue; the resulting high peak OP margin is a deliberate operating-leverage assumption requiring verification, not a reported target.

Profit basis: GAAP-style consolidated OP including SBC; ceiling uses the parent economic OP proxy shown separately. Amounts below are USD millions.

OP = sum of product GP/contribution - explicit operating costs. OP margin = OP ÷ revenue. Displayed numbers are rounded for reading; the workbook calculates from unrounded inputs. Guidance assigned to a future quarter retains Model status because guidance is not an actual result.

Parent economic operating-profit bridge

Inherited reference model. Tongmei noncontrolling/redeemable interests total 14%, leaving an 86% parent claim. Apply 86% to consolidated operations before parent-only expenses, then subtract those expenses in full. Parent costs already in opex are estimated at 2M per quarter in H2 2026 and $2,500,000 / $3,000,000 / $3,500,000 / $4,000,000 in 2027-30. The approximation does not separately allocate lower-tier raw-material minorities and therefore can still overstate parent capture. No speculative Hong Kong IPO share count is invented: a further 10% dilution of the subsidiary stake would reduce the operating claim by approximately 10%, slightly more after fixed parent costs. Replace this proxy when completed terms and segment costs are known.

Attributable OP proxy = (consolidated OP + parent-only costs) × subsidiary ownership - parent-only costs. The current ceiling chapter uses this proxy and distinguishes consolidated from parent economic OP.

8. Multiple, shares and recognition convention

Keep 20× forward four-quarter parent economic OP. The 86% ownership allocation is not an extra risk-probability discount. No subsidiary market capitalization or cash from a future IPO is added to the operating ceiling.

Inherited reference model. Use 66,500,000 diluted shares for H2 2026, matching Q3 guidance, then 68/70/72/74M in 2027-30 to allow continuing awards and capital needs. The completed April offering is already reflected in the current denominator; it is not charged again. June issued common shares were about 65,570,000 versus the Q2 weighted-average diluted 63,474,000. Authorized-share capacity is not issued dilution. Tongmei financing affects ownership separately from parent shares; no announced final Hong Kong listing terms were verified.

The valuation denominator is the ceiling quarter’s diluted-share proxy. It is not a historical EPS weighted-average assumption disguised as a fully known future capitalization. Unit factor: 1; this converts the model units to USD per share. No cash balance, enterprise-value adjustment or after-tax earnings substitution is added to the operating-profit ceiling.

These are calendar-year sums of the quarterly path, not extra revenue and not the live valuation method. For non-December fiscal reporters, fiscal guidance and these calendar totals must be aligned using the period mapping above; a four-quarter run rate is not automatically the same fiscal year. Outer-year outputs are conditional TAM/capture trajectories, not verified management sales targets.

The reference capacity plan reaches USD130M quarterly InP revenue capacity at the end of 2027. Revenue differs only by the explicit policy price/displacement cases. Capacity is not annual sales. The 2028-30 tail still requires additional qualified capacity and export permissions; the six-inch mix is inside total InP, not additive to it.

12. Programme and probability ledger

Programme / signal	Source and probability	Amount / unit base	Treatment in this model	Escalation / recognition
Casela	Signed; 90% delivery judgment	RMB173M, approximately 25,400,000 USD, for 2027	Monthly deliveries inside InP sales; 50% prepay due initially and 50% by year-end 2026	Contract value is not proof that all cash was already received.
Coherent	Signed; 85% performance judgment	22,288,500 prepayment; three-year capacity agreement	Deposit is a liability until product ships; within InP curve	Unused amounts can be refundable; >6-month capacity miss permits termination.
Lumentum	Signed; 85% performance judgment	43,500,000 initial deposit and 43,500,000 second deposit with terms to be set during 2028	Reservation/credits, not 87M of extra sales; total purchases undisclosed	Confirm actual receipt and delivery; supplier failure can require unused-deposit refund.
2027 InP capacity	Management plan; 75% execution judgment	About 130M quarterly revenue capacity exiting 2027	Model reaches that level in Q4 only	Validate yield, diameter mix and permits.
Tongmei listing	Proposed Hong Kong route; 60% timing judgment	STAR application withdrawn; prior investors have redemption rights	No IPO proceeds or issued shares presumed	49M original PE investment may be redeemed; parent had funding capacity at filing.

Probabilities are analyst judgments under the framework signal hierarchy. Embedded revenue is not multiplied or added a second time. Where no defensible amount or unit base exists, the opportunity remains explicitly unquantified; confidence alone cannot create a revenue number. New evidence can enlarge the TAM, improve capture or move recognition earlier, but each requires an identifiable model input.

13. Scored risk classes

Risk class	Scored deduction	Mechanism and consequence
Export permits and geopolitics	-1	Order backlog cannot substitute for permission to ship to a destination.
Scale, quality and investment execution	-1	Larger crystals and adjacent facilities may miss yield or delivery schedules.
Customer obligations and demand duplication	-1	Prepayments can require refunds; customers may reserve buffers across suppliers.
Competition and optical substitution	-1	Rival substrate expansion, VCSEL and alternate architectures can reduce scarcity and pricing.
Ownership, redemption and dilution	-1	Tongmei minority claims, potential IPO and parent awards change shareholder economics.

14. Dated catalysts, counters and escalation

1. **Q3 2026 results - Permit-covered revenue** Test 66M sales, 11M GAAP opex, cash conversion and updated permits.
2. **Q4 2026 - InP capacity** Separate productive 60M exit capacity from shipments and signed orders.
3. **2027 - New facility and contract deliveries** Casela schedule, qualification and 130M exit InP capacity are concrete conversion tests.
4. **2027-28 - Hong Kong listing or redemption** Replace ownership proxy with completed terms; no automatic benefit from planned listing.
5. **Each quarter - Diameter disclosure and margins** A disclosed 6-inch split, wafer-equivalent units and actual margins would materially improve confidence.
6. **2028-30 - Capacity versus demand** Lower ASP, customer buffers or rival capacity require revising the shortage premium, not assuming a permanent margin floor.

15. Cross-book monitoring

Lumentum and Coherent are verified counterparties; their chip-fab expansion can need merchant substrates even when they integrate more device production. The old assertion that Coherent device capacity necessarily competes directly with AXT crystal substrates is too broad. AAOI and other transceiver/laser suppliers sit downstream. Optical growth benefits multiple stages but their revenues and margins are not additive in one company model.

16. Source register and model limits

S1 - Issuer Q2 results (2026-07-30). GAAP results, diluted shares and balance sheet.

S2 - Q2 Form 10-Q (2026-08-13). Ownership, category reconciliation, cash flow and contract terms.

S3 - Q1 management call (2026-04-30). Full displayed speaker sequence reviewed through closing.

S4 - Q2 management call (2026-07-30). Full displayed speaker sequence reviewed through closing.

S5 - SEC-filed earnings release (2026-07-30). Primary financial cross-check against public-call transcription.

Product margins, diameter mix and post-2027 capacity remain LOW-confidence assumptions. The old claim that a 54%+ InP margin was already proved by consolidated GM is removed: many segment combinations can reconcile one blended result. No independently verified exact global supplier shares or wafers-per-month capacity is used. Fixed 20× OP is not a PE multiple. Interest earned on offering proceeds is below OP and excluded.

Complete current operating model

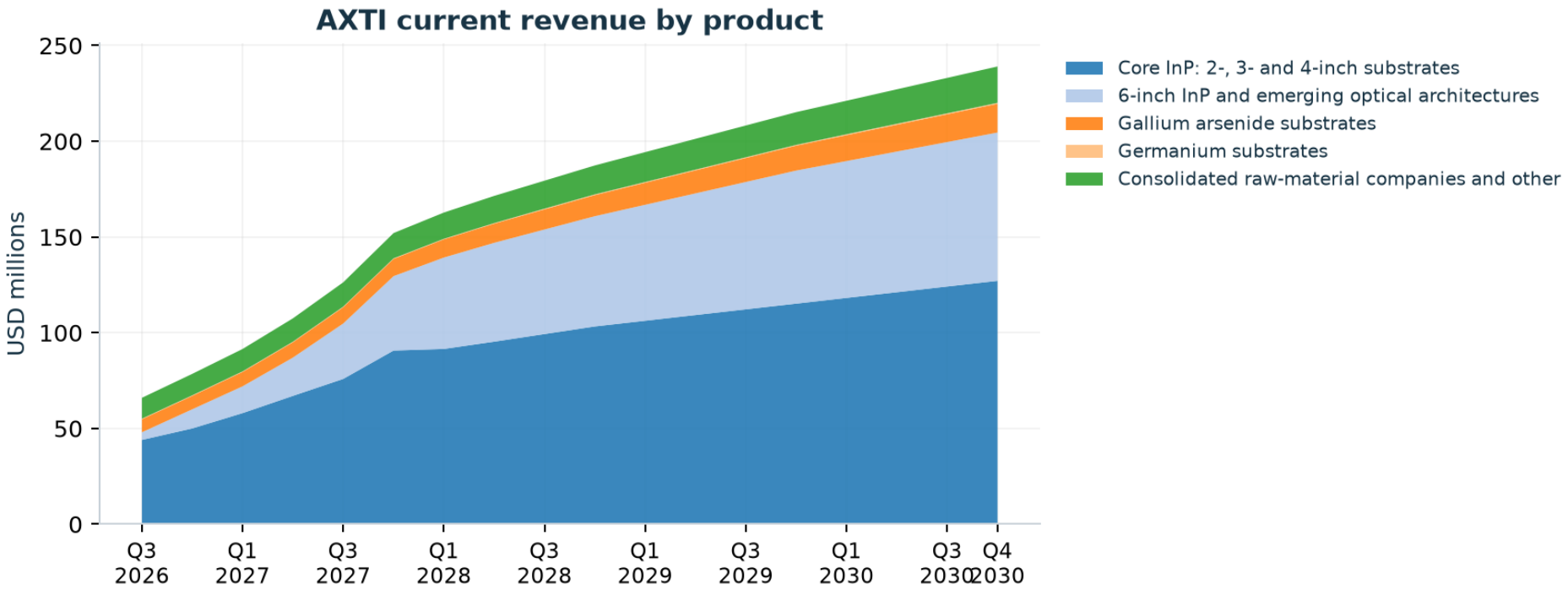


Figure 4. Current quarterly revenue by product, USD millions. All inherited product vectors remain; AAOI alone adds a separately labeled substitution vector.

Quarter	Revenue, USD thousands	Gross profit, USD thousands	COGS, USD thousands
Q3 2026	66,000	29,871	36,129
Q4 2026	78,500	38,867	39,633
Q1 2027	91,500	46,752	44,748
Q2 2027	107,500	56,532	50,968
Q3 2027	126,316	68,057	58,260
Q4 2027	152,045	83,382	68,663
Q1 2028	162,765	89,395	73,370
Q2 2028	171,464	92,545	78,919

Quarter	Revenue, USD thousands	Gross profit, USD thousands	COGS, USD thousands
Q3 2028	179,415	94,870	84,545
Q4 2028	187,366	96,785	90,581
Q1 2029	194,324	98,670	95,654
Q2 2029	201,282	100,520	100,762
Q3 2029	208,240	102,251	105,989
Q4 2029	215,198	104,742	110,456
Q1 2030	221,163	106,564	114,599
Q2 2030	227,128	109,445	117,683
Q3 2030	233,093	111,339	121,754
Q4 2030	239,058	114,183	124,875

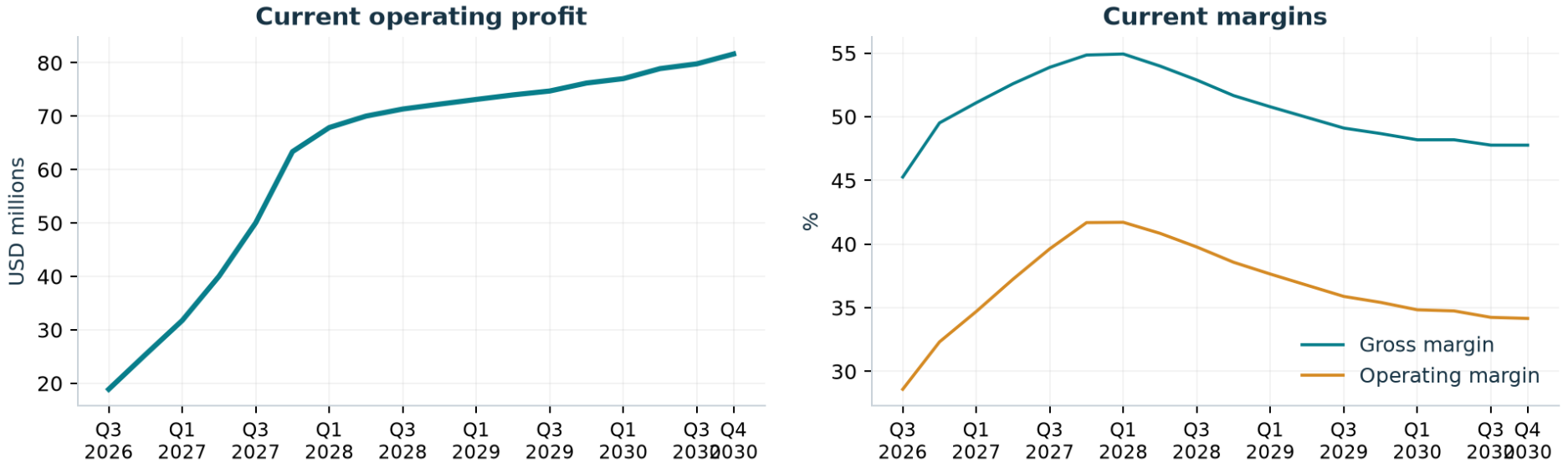


Figure 5. Current operating profit and margins. Operating profit is consolidated USD millions; the valuation uses parent-economic OP only where the inherited method already requires it.

Quarter	Revenue, USD thousands	Gross margin	Opex, USD thousands	Operating profit, USD thousands	OP margin	Valuation OP, USD thousands
Q3 2026	66,000	45%	11,000	18,871	29%	15,845
Q4 2026	78,500	50%	13,500	25,367	32%	21,399
Q1 2027	91,500	51%	15,000	31,752	35%	26,786
Q2 2027	107,500	53%	16,500	40,032	37%	33,865
Q3 2027	126,316	54%	18,004	50,053	40%	42,433
Q4 2027	152,045	55%	20,010	63,373	42%	53,821
Q1 2028	162,765	55%	21,516	67,879	42%	57,602
Q2 2028	171,464	54%	22,522	70,023	41%	59,435
Q3 2028	179,415	53%	23,523	71,347	40%	60,567
Q4 2028	187,366	52%	24,524	72,261	39%	61,348
Q1 2029	194,324	51%	25,525	73,144	38%	62,031
Q2 2029	201,282	50%	26,526	73,994	37%	62,757
Q3 2029	208,240	49%	27,527	74,724	36%	63,382
Q4 2029	215,198	49%	28,528	76,214	35%	64,655
Q1 2030	221,163	48%	29,529	77,036	35%	65,286
Q2 2030	227,128	48%	30,529	78,915	35%	66,893
Q3 2030	233,093	48%	31,530	79,809	34%	67,656
Q4 2030	239,058	48%	32,531	81,652	34%	69,233

Every revenue vector and product margin

Product GP = product revenue x the displayed modeled margin. Consolidated gross profit adds the unchanged source-model adjustment where applicable (including Sivers Glasgow depreciation). Product allocations and future margins remain LOW-confidence model assumptions unless specifically identified as historical or guided.

Core InP: 2-, 3- and 4-inch substrates

Quarter	Revenue, USD thousands	Gross margin	Gross profit, USD thousands
Q3 2026	44,000	52%	22,660
Q4 2026	50,000	56%	27,750
Q1 2027	58,000	56%	32,770
Q2 2027	67,000	57%	38,391
Q3 2027	75,867	58%	43,864
Q4 2027	90,682	58%	52,445
Q1 2028	91,517	58%	52,671
Q2 2028	95,328	56%	53,644
Q3 2028	99,300	55%	54,492
Q4 2028	103,272	53%	55,126
Q1 2029	106,251	52%	55,655
Q2 2029	109,230	51%	56,126
Q3 2029	112,209	50%	56,536
Q4 2029	115,188	50%	57,463
Q1 2030	118,167	49%	58,359
Q2 2030	121,146	49%	59,830
Q3 2030	124,125	49%	60,682
Q4 2030	127,104	49%	62,139

6-inch InP and emerging optical architectures

Quarter	Revenue, USD thousands	Gross margin	Gross profit, USD thousands
Q3 2026	4,000	56%	2,240
Q4 2026	10,000	60%	5,950
Q1 2027	14,000	60%	8,470
Q2 2027	20,000	62%	12,300
Q3 2027	28,949	62%	18,011
Q4 2027	38,864	63%	24,418
Q1 2028	47,748	63%	29,864
Q2 2028	51,636	61%	31,685
Q3 2028	54,615	60%	32,805
Q4 2028	57,594	59%	33,732
Q1 2029	60,573	57%	34,751
Q2 2029	63,552	56%	35,826
Q3 2029	66,531	55%	36,841
Q4 2029	69,510	55%	38,075
Q1 2030	71,496	54%	38,663
Q2 2030	73,482	54%	39,737
Q3 2030	75,468	54%	40,435
Q4 2030	77,454	54%	41,499

Gallium arsenide substrates

Quarter	Revenue, USD thousands	Gross margin	Gross profit, USD thousands
Q3 2026	6,800	30%	2,040
Q4 2026	7,000	30%	2,135
Q1 2027	7,500	31%	2,325
Q2 2027	8,000	31%	2,496
Q3 2027	8,500	32%	2,678
Q4 2027	9,000	32%	2,880
Q1 2028	9,500	32%	3,059
Q2 2028	10,000	32%	3,250
Q3 2028	10,500	33%	3,444
Q4 2028	11,000	33%	3,630
Q1 2029	11,500	33%	3,795
Q2 2029	12,000	33%	3,960
Q3 2029	12,500	33%	4,125
Q4 2029	13,000	33%	4,316
Q1 2030	13,500	33%	4,482
Q2 2030	14,000	33%	4,676
Q3 2030	14,500	33%	4,843
Q4 2030	15,000	34%	5,025

Germanium substrates

Quarter	Revenue, USD thousands	Gross margin	Gross profit, USD thousands
Q3 2026	300	25%	75
Q4 2026	300	25%	75
Q1 2027	300	25%	75
Q2 2027	300	25%	75
Q3 2027	300	25%	75
Q4 2027	300	25%	75
Q1 2028	300	25%	75
Q2 2028	300	25%	75
Q3 2028	400	25%	100
Q4 2028	400	25%	100
Q1 2029	400	25%	100
Q2 2029	400	25%	100
Q3 2029	400	25%	100
Q4 2029	400	25%	100
Q1 2030	500	25%	125
Q2 2030	500	25%	125
Q3 2030	500	25%	125
Q4 2030	500	25%	125

Consolidated raw-material companies and other

Quarter	Revenue, USD thousands	Gross margin	Gross profit, USD thousands
Q3 2026	10,900	26%	2,856
Q4 2026	11,200	26%	2,957
Q1 2027	11,700	27%	3,112
Q2 2027	12,200	27%	3,270
Q3 2027	12,700	27%	3,429
Q4 2027	13,200	27%	3,564
Q1 2028	13,700	27%	3,726
Q2 2028	14,200	27%	3,891
Q3 2028	14,600	28%	4,030
Q4 2028	15,100	28%	4,198
Q1 2029	15,600	28%	4,368
Q2 2029	16,100	28%	4,508
Q3 2029	16,600	28%	4,648
Q4 2029	17,100	28%	4,788
Q1 2030	17,500	28%	4,935
Q2 2030	18,000	28%	5,076
Q3 2030	18,500	28%	5,254
Q4 2030	19,000	28%	5,396

Full ceiling derivation

Valuation OP equals consolidated OP except where the saved model allocates economics to parent shareholders. The current model adds ownership x incremental OP to the saved parent-economic OP. Existing parent-only expenses and senior claims remain deducted as before. For every ceiling, sum that quarter and its next three quarters, multiply by the sanctioned multiple and currency/unit factor, and divide by the saved quarter-specific share denominator. No stock-price-dependent upside or ranking appears in this artifact.

Ceiling quarter / window	F4Q valuation OP, USD thousands	Multiple	Shares	FX / unit factor	Ceiling
Q3 2026 to Q2 2027	97,893	20x	66,500,000	1	29
Q4 2026 to Q3 2027	124,481	20x	66,500,000	1	37
Q1 2027 to Q4 2027	156,904	20x	68,000,000	1	46
Q2 2027 to Q1 2028	187,720	20x	68,000,000	1	55
Q3 2027 to Q2 2028	213,290	20x	68,000,000	1	63
Q4 2027 to Q3 2028	231,424	20x	68,000,000	1	68
Q1 2028 to Q4 2028	238,951	20x	70,000,000	1	68
Q2 2028 to Q1 2029	243,380	20x	70,000,000	1	70
Q3 2028 to Q2 2029	246,703	20x	70,000,000	1	70
Q4 2028 to Q3 2029	249,518	20x	70,000,000	1	71
Q1 2029 to Q4 2029	252,825	20x	72,000,000	1	70
Q2 2029 to Q1 2030	256,079	20x	72,000,000	1	71
Q3 2029 to Q2 2030	260,215	20x	72,000,000	1	72
Q4 2029 to Q3 2030	264,490	20x	72,000,000	1	73
Q1 2030 to Q4 2030	269,067	20x	74,000,000	1	73

Q4 2027 uses the saved four-quarter parent operating profit, multiplied by 20 and divided by 68 million starting-quarter shares. The current ceiling is about USD 68; exact inputs remain in the full model.

Annual cross-check

Calendar-year sums below include only the quarters in the saved model. Partial years are labeled. These are not additional revenue, a horizon roll, or new management guidance.

Calendar year	Coverage	Revenue, USD thousands	Gross profit, USD thousands	Operating profit, USD thousands	OP margin
2026	2 quarters	144,500	68,738	44,238	31%
2027	4 quarters	477,361	254,723	185,209	39%
2028	4 quarters	701,010	373,596	281,510	40%

Calendar year	Coverage	Revenue, USD thousands	Gross profit, USD thousands	Operating profit, USD thousands	OP margin
2029	4 quarters	819,044	406,182	298,076	36%
2030	4 quarters	920,442	441,531	317,412	34%

Probability and dilution sensitivity

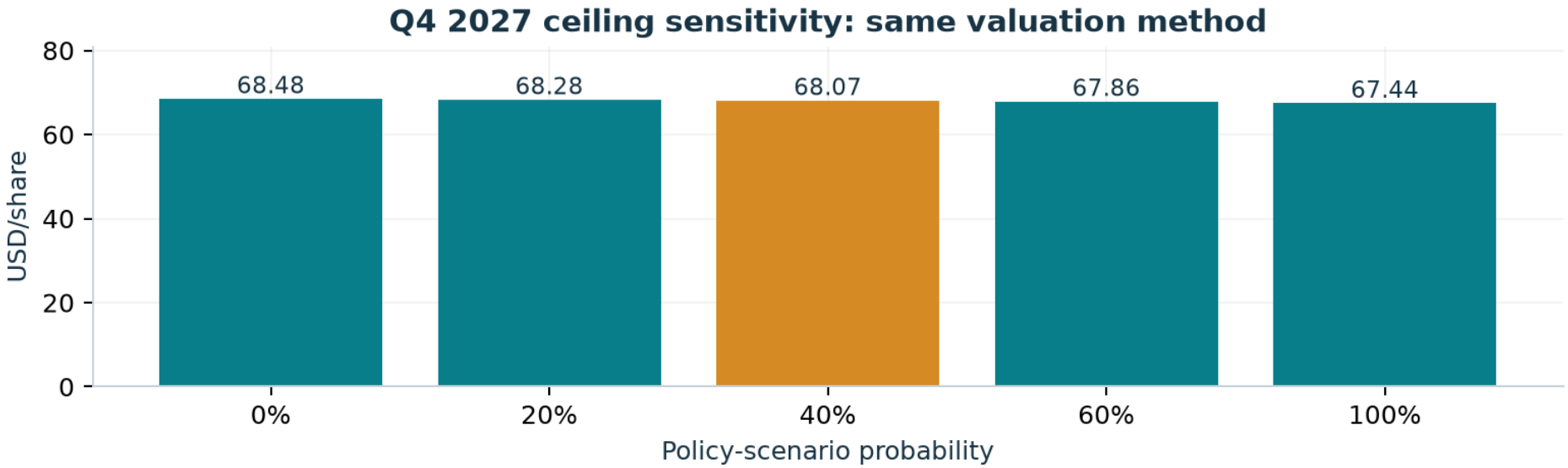


Figure 6. Q4 2027 ceiling under alternative policy probabilities. Gold is the selected 40% weight. All company valuation methods, operating reference inputs and shares remain unchanged.

Policy probability	Q4 2027 ceiling	Change from zero-policy sensitivity
0%	68	0%
20%	68	-31 basis points
40%	68	-61 basis points
60%	68	-92 basis points
100%	67	-2%

Additional diluted shares	Q4 2027 Ceiling	Interpretation
0%	68	Same operating case; denominator sensitivity only
5%	65	Same operating case; denominator sensitivity only
10%	62	Same operating case; denominator sensitivity only

The 40% weight is a judgment about the incremental policy economics and timing, not a statistical certainty. Because the nine companies share the same policy event, these increments should not be treated as nine independent sources of portfolio diversification. The next 12-16 months have the highest forecasting relevance; later continuation assumptions are progressively less reliable.

Validation and next evidence

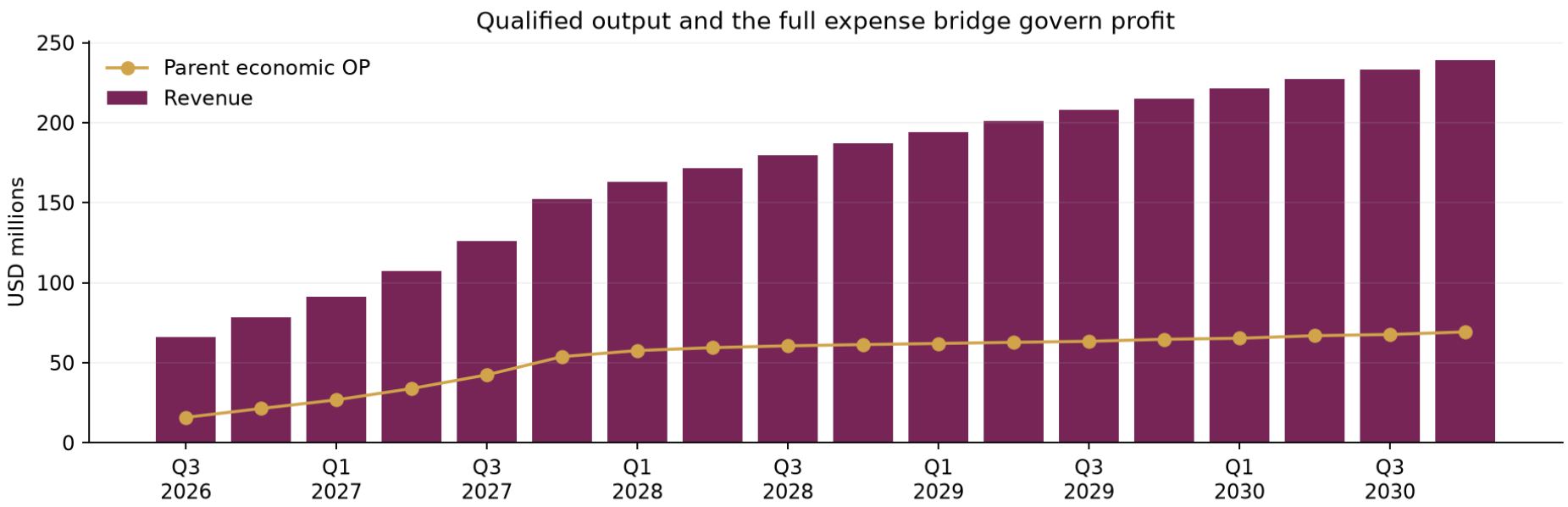
This current artifact and the master model use the same selected 40% policy case. Existing company valuation methods, quarter conventions, ownership and dilution schedules are retained. Every existing portfolio Excel Base is protected; the analytical score above is assessed separately.

The incremental ledger exposes revenue, production costs, gross profit and operating expense. A zero-policy sensitivity removes only that ledger's contribution. It is a diagnostic control, not a competing active forecast. Reported facts and dated guidance retain their original dates; new pricing, exposure, costs and timing are labeled analyst assumptions.

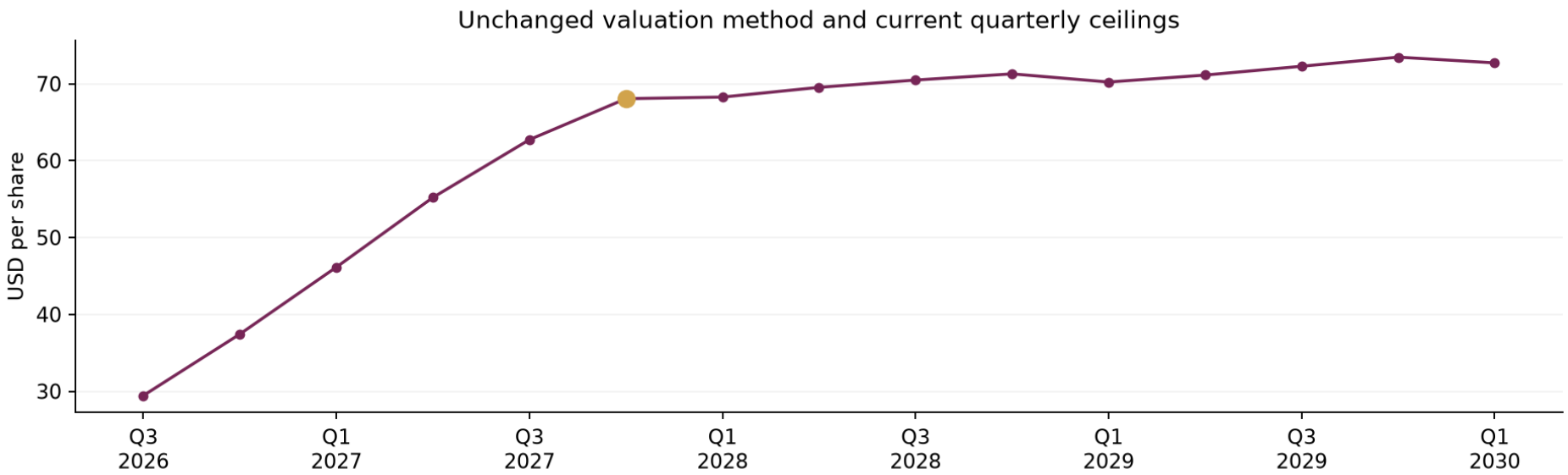
The most useful next evidence is the proposed content definition, product-specific fabrication and assembly routes, contract repricing rights, delivered 3.2T qualification, accepted component premiums and unit-cost/yield data. Those observations can raise or lower the inputs and their probability; official confirmation is not a prerequisite for modeling this credible breadcrumb.

Current model: **AXTI_Q_Model** in **v3_2_master_portfolio.xlsx**. The full quarterly model includes all original product vectors and the approved incremental operating bridge. Approval snapshots and change history are retained externally rather than as competing current forecasts.

Current full model visual checks



Current revenue and full operating-profit path; financial inputs are unchanged.



Current ceiling path. Gold marks Q4 2027; methods, earnings windows and share conventions are retained.